



North South University

Department of Mathematics and Physics

Instructor: Dr. M Abdur Rab(AuR)

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Semester : Fall 2026

Office : SAC1050

Office time: 11:30AM - 1:30PM(MW)

Module : MAT 125: Introduction to Linear Algebra

Coordinator : Dr. Md. Monir Uddin (monir.uddin@northsouth.edu)

Text Book : Elementary Linear Algebra By Howard Anton (11th Edition)
supplementary : Introduction to Linea Algebra By Gilbert Strang (Third Edition)

Credit : 3 credit points

Marks Distribution :	Midterm	30%
	Quizzes	20%
	Attendance	5%
	Assignment	5%
	Assessment of Assignment	5%
	<u>Final Exam</u>	<u>35%</u>
	Total	100%

Contents

Mid-Term Exam

Chapter 1: System of Linear Equations and Matrices

- Introduction to System of Linear Equations
- Gaussian Eliminations
- Matrices and Matrix Operations
- Inverse; Rules of Matrix Arithmetic
- Elementary Matrices and a Method for Finding inverse of Matrix
- Further Results on Systems of Equations and Invertibility
- Diagonal, Triangular and Symmetric Matrices

Chapter 2: Determinants:

- Determinant by Cofactor Expansion
- Evaluating Determinants by Row Reduction

- Properties of Determinant Function

Chapter 3. Euclidean Vector Spaces:

- Euclidean n -space
- Linear Transformation from $\mathbf{R}_n$ to $\mathbf{R}_m$
- Properties of Linear Transformations

Final Exam

Chapter 4: General Vector Spaces

- Real Vector Spaces
- Subspaces
- Linear Independence and Dependence
- Basis, Dimension, Solution Space and Null Space
- Row Space, Column Space and Null Space
- Rank and Nullity

Chapter 5: Eigenvalues and Eigenvectors

- Eigenvalues and Eigenvectors
- Diagonalization

Chapter 10: Applications of Linear Algebra

- Electrical Net works
- Linear Programming
- Graph Theory

Suggest only for reading: (Out of lectures)

Chapter 3: Vectors in 2-Space and 3-Space

- Introduction to Vectors
- Norm of a Vector; Vector Arithmetic
- Dot Product; Projections
- Lines and Planes in 3-Space

Chapter 6: Inner Product Spaces

- Inner Products
- Angle and Orthogonality in Inner Products

- Orthonormal Bases; Gram-Schmidt Process
- Orthogonal Matrices; Change of Basis